

Host guide: Smart-RUDP review

For the presenter only. Each page shows one slide of the real deck, what to say, and the question it is likely to bring.

Computer Networks | Faculty: Anuradha G | 9 Oct 2026

Plan for the session

Total talking time about 15 minutes, then questions.

#	Slide	Time	#	Slide	Time
1	Title	0:20	15	Smart-RUDP design	0:50
2	The problem	0:40	16	The AI agent in our code	1:00
3	Why we built it	0:50	17	Smart-RUDP result	0:50
4	What we built	0:40	18	Wireshark traces	0:40
5	Packet header	0:40	19	Wireshark real capture	0:50
6	Handshake and data flow	0:50	20	RUDP Chat demo	1:00
7	Go-Back-N	0:50	21	More evidence: baselines, SACK, reordering	1:00
8	AIMD	0:50	22	Neural agent in the C transport	1:00
9	How we tested	0:40	23	Fairness, ablations, chat fixes	0:50
10	Benchmark results	0:40	24	Safety net: hand over to AIMD	0:50
11	Problems fixed	0:40	25	Selective Repeat is now the default	0:50
12	Literature	0:30	26	Learnings and future work	0:50
13	Network design	0:50	27	Thank you	0:15
14	Packet Tracer results	0:40	28	Appendix: slow start	-

Rules: say the limits before she asks; answer in one or two sentences; if you do not know, say what you would check.

1. Title

Time on this slide: 0:20

Smart-RUDP

Reliable UDP with learning-based congestion control, in C

Computer Networks | Faculty: Anuradha G

Harshavardhan Mallela 24BAI0274 Rohit Arya 24BCE0819 Shloak Sinha 24BDS0378

SAY

- Good morning ma'am. We are Harshavardhan, Rohit and Shloak.
- Our project is Smart-RUDP: a reliable transport protocol we built in C on top of UDP, plus a controller that learns how fast to send.

IF ASKED Say the three names and the GitHub link once, then move on.

2. The problem

Time on this slide: 0:40

The problem

UDP

Fast and simple, but no guarantees: packets can be lost, duplicated or reordered.

TCP

Reliable, but lives inside the OS kernel. Hard to see, hard to change, hard to learn from.

In 1986 the early Internet suffered congestion collapse: throughput fell from 32 Kbps to 40 bps. TCP congestion control was invented to fix it.

Our question: can we rebuild TCP's core ideas ourselves, on top of UDP?

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SAY

- UDP is fast but gives no guarantees: packets can be lost, duplicated or reordered.
- TCP is reliable but sits inside the operating system, so you cannot see or change it.
- In 1986 the early Internet had congestion collapse: throughput fell from 32 Kbps to 40 bps. TCP's congestion control was invented to fix that.
- Our question: can we rebuild TCP's core ideas ourselves, on UDP?

IF ASKED Q: Why not just use TCP? A: TCP cannot be changed or inspected; building our own makes every mechanism visible. QUIC and HTTP/3 do the same in practice.

3. Why we built it

Time on this slide: 0:50

Why we built it

The need

Campus users send data to a server across a routed network. TCP is fixed inside the kernel; plain UDP gives no guarantees. Real systems (QUIC, HTTP/3) solve this with reliability built in user space over UDP.

Our answer

RUDP: a readable, tunable reliable transport over UDP, plus Smart-RUDP, a controller that learns the window from RTT and loss.

Scenario: a student on VLAN 10 uploads a file to the server LAN. The ACL keeps students out of staff VLAN 20. If a core link fails, OSPF reroutes through R2: a longer path and higher RTT that the transport must adapt to.

Validated: protocol tests and Smart-RUDP vs AIMD on emulated paths; network design built and tested in Packet Tracer; traffic analysed in Wireshark.

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SAY

- The need: campus users send data to a server across a routed network. TCP is fixed in the kernel; UDP gives no guarantees.
- Our answer: RUDP, a readable and tunable transport, and Smart-RUDP, which learns the window from RTT and loss.
- Scenario: a student on VLAN 10 uploads to the server LAN. The ACL keeps students out of staff VLAN 20. If a core link fails, OSPF reroutes through R2: longer path, higher RTT, and the transport must adapt.

IF ASKED Q: Is this just a lab exercise? A: A lab sends datagrams. We built reliability and congestion control on top, measured them, and used them in an application.

4. What we built

Time on this slide: 0:40

What we built

Handshake

SYN, SYN-ACK, ACK
before any data flows.

Reliability

Sequence numbers,
ACKs, checksums,
timeouts,
retransmission.

Flow control

Go-Back-N sliding
window; its size is the
congestion window.

Congestion control

AIMD: slow start,
avoidance, halve on
loss.

About 670 lines of C across 4 source files. Client + server, built in 8 phases, on github.com/Capsturrrr/rudp

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SAY

- Four pieces: handshake, reliability, flow control and congestion control.
- About 670 lines of C across four source files, built in eight phases, on GitHub at [Capsturrrr/rudp](https://github.com/Capsturrrr/rudp).

IF ASKED Q: Who did what? A: Agree one answer as a team before you go in.

5. Packet header

Time on this slide: 0:40

The RUDP packet header: 13 bytes

Field	Size	Purpose
seq	4 B	Packet number
ack	4 B	Next packet expected
flags	1 B	SYN / ACK / FIN / DATA
checksum	2 B	Detects corruption
payload_len	2 B	Data length (max 512)

Fields packed byte by byte, so compiler struct padding never changes the wire format.

Data packet = 53 bytes on the wire. ACK / handshake = 13 bytes.

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SAY

- 13-byte header: sequence 4, ack 4, flags 1, checksum 2, payload length 2.
- Fields are packed byte by byte so compiler padding never changes the wire format.
- A data packet is 53 bytes on the wire; ACKs and handshake packets are 13.

IF ASKED Q: How does the checksum work? A: 16-bit one's-complement sum over header and payload, like UDP/TCP. A corrupted packet fails and is recovered by retransmission.

6. Handshake and data flow

Time on this slide: 0:50

Handshake and data flow

1	SYN	Client -> Server	Request a connection
2	SYN-ACK	Client <- Server	Server agrees
3	ACK	Client -> Server	ESTABLISHED, data may flow
4	DATA	Client -> Server	Up to window-size packets in flight
5	ACK	Client <- Server	Cumulative: confirms everything up to N
6	FIN	Client -> Server	Close the connection

Server accepts a new SYN in any state, so a killed client can never leave it stuck.

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SAY

- Three-way handshake: SYN, SYN-ACK, ACK. Then data, up to a window of packets in flight.
- ACKs are cumulative: ACK N confirms everything before N. FIN closes the connection.
- The server accepts a new SYN in any state, so a killed client can never leave it stuck. That was a real bug we fixed.

IF ASKED Q: What if the peer disappears? A: Sender keeps retransmitting on timeout. An abort after repeated timeouts is not implemented; it is the fix we would add.

7. Go-Back-N

Time on this slide: 0:50

Reliability: Go-Back-N

- Window = cwnd packets in flight (fixed at 4 in Phase 4, then driven by cwnd)
- Receiver sends cumulative ACKs
- Retransmission timeout: 500 ms
- Fast retransmit after 3 duplicate ACKs, no waiting for the timer
- Checksum failures are dropped, then recovered by retransmission

Proof

In a netem 20% loss capture we saw 12 timeouts spaced about 500 ms apart.

Result

2,442-byte message reassembled correctly.

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SAY

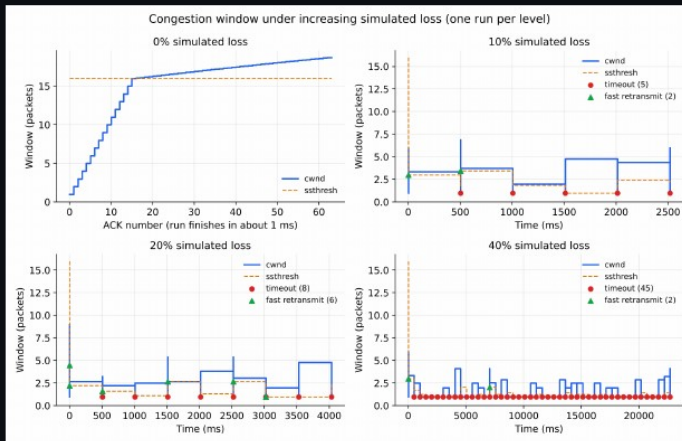
- The window is the number of packets in flight; it started fixed at 4 and is now driven by the congestion window.
- Timeout is 500 ms; fast retransmit triggers after 3 duplicate ACKs, no waiting for the timer.
- Proof: in a 20% loss netem capture we saw 12 timeouts about 500 ms apart. Result: the 2,442-byte message reassembled correctly.

IF ASKED Q: Why Go-Back-N? A: Simpler to build and explain. It wastes bandwidth under loss; Selective Repeat with SACK is our stated future work.

8. AIMD

Time on this slide: 0:50

Congestion control: AIMD



Fresh runs at 0%, 10%, 20% and 40% simulated loss. Red = timeout, green = fast retransmit.

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SAY

- Slow start: window grows by one per ACK until the threshold of 16. Then avoidance: plus $1/\text{window}$ per ACK.
- Three duplicate ACKs: window halves. A timeout: window resets to 1 and the threshold halves. Cap is 32.
- Timeout: cwnd resets to 1, ssthresh halves
- Capped at 32
- The graph shows fresh runs at 0, 10, 20 and 40 percent loss. Red marks are timeouts, green are fast retransmits.

IF ASKED Q: Why cap 32 and ssthresh 16? A: The cap stops a lossless run growing without bound; 16 is where doubling switches to linear growth.

9. How we tested

Time on this slide: 0:40

How we tested

Built-in loss simulator

Drops packets in code at a set percentage. Used for the benchmark.

tc / netem

Linux kernel injects real loss on loopback. Used to validate behaviour.

Wireshark / tshark

Captured in WSL, opened in Windows Wireshark. Confirmed handshake, ACKs, retransmissions.

Wireshark could not see WSL loopback from Windows, so we captured with tshark inside WSL and copied the file across.

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SAY

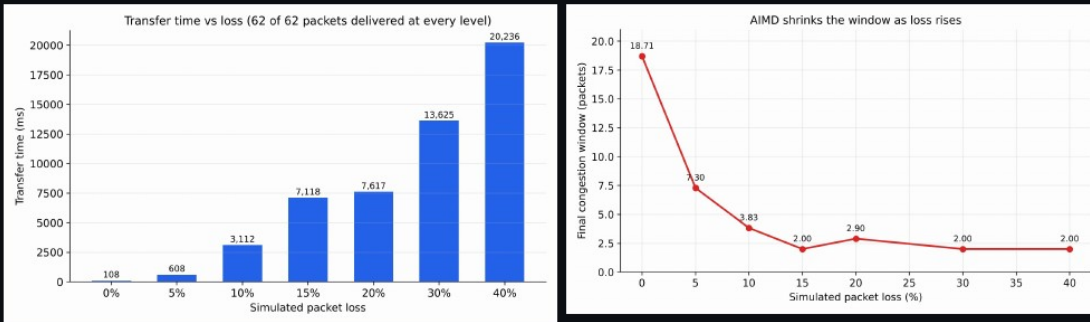
- Three tools: a loss simulator inside the code (used for the benchmark), Linux tc/netem injecting real loss on loopback, and Wireshark/tshark.
- Windows Wireshark could not see WSL loopback, so we captured with tshark inside WSL and copied the file across.

IF ASKED Q: Is the loss real? A: Benchmark and chat loss is simulated in the program. We also captured a real loopback run with tshark and netem loss.

10. Benchmark results

Time on this slide: 0:40

Benchmark results



62 of 62 packets delivered at every loss level, from 0% to 40%.

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SAY

- 62 of 62 packets were delivered at every loss level from 0 to 40 percent.
- Higher loss costs time and retransmissions, never correctness.

IF ASKED Q: Single runs? A: Yes, the original benchmark ran each level once. Smart-RUDP evaluation uses 6 runs per scenario.

11. Problems fixed

Time on this slide: 0:40

Problems we hit and fixed

Problem	Fix
Server stuck after killed client	Accept SYN in any state (was a 15.5 s hang)
Leftover netem rule	Check tc qdisc show dev lo, delete before each test
Windows Wireshark saw nothing	Capture with tshark inside WSL, then copy
Shell portability in benchmark	Replaced date/grep tricks with awk and sed

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SAY

- Server stuck after a killed client: it hung 15.5 s; fix was to accept SYN in any state.
- A leftover netem rule distorted results; we now check tc qdisc before each test.
- Windows Wireshark saw nothing: capture inside WSL.
- Benchmark script portability: replaced date and grep tricks with awk and sed.

IF ASKED Say these quickly: it shows real debugging, which is what separates a project from a lab.

12. Literature

Time on this slide: 0:30

Literature behind the design

- Jacobson & Karels (1988): congestion avoidance and control, the origin of AIMD
- Chiu & Jain (1989): why additive increase, multiplicative decrease converges fairly
- RFC 5681 (2009): TCP congestion control standard
- Ha, Rhee, Xu (2008): CUBIC, our future-work direction
- Langley et al. (2017) and RFC 9000: QUIC, reliable transport over UDP in production
- Midhula & Arun Raj Kumar (2024) and Cao et al. (2023, DRLFcc): deep reinforcement learning for congestion control, the idea behind Smart-RUDP

Key researcher: Van Jacobson

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SAY

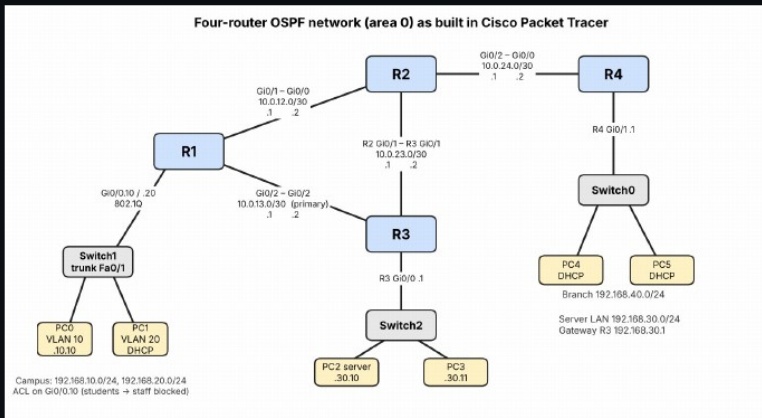
- Jacobson and Karels 1988 is the origin of AIMD; Chiu and Jain 1989 explains why it converges fairly.
- RFC 5681 is the standard; CUBIC is future work; QUIC and RFC 9000 are reliable transport over UDP in production.
- The AI part comes from Midhula and Arun Raj Kumar (2024) and Cao et al. (2023, DRLFcc): deep reinforcement learning for congestion control. Smart-RUDP applies the same idea with a smaller tabular Q-learning agent.
- Key researcher: Van Jacobson.

IF ASKED Q: Why AIMD? A: Chiu and Jain show it converges to a fair share. Q: Are those papers deep RL and yours is not? A: Yes; we used a lookup table, which is simpler and inspectable.

13. Network design

Time on this slide: 0:50

Network design: OSPF topology in Packet Tracer



Built and tested in Packet Tracer: OSPF adjacencies, routes, ACL and failover all verified.

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SAY

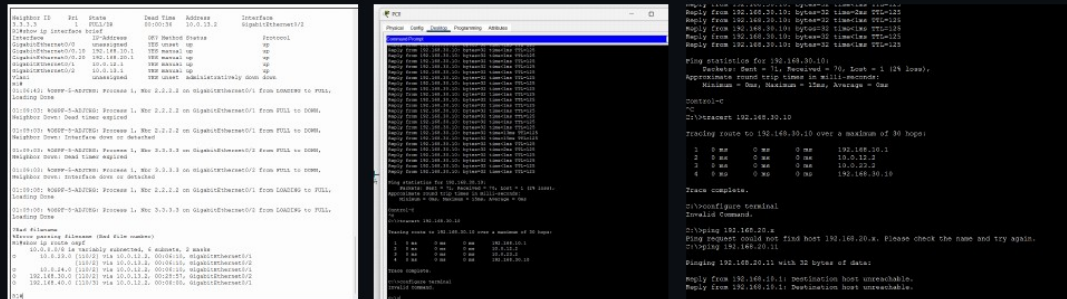
- Four routers, OSPF area 0: R1, R2, R3 form a triangle and R4 is a branch.
- Two VLANs with router-on-a-stick, DHCP on R1 and R4, and an access list blocking student to staff.
- Routes: server LAN metric 2 via R3, branch metric 3 via R2.
- Failover: shutting R1 Gi0/2 still gave 70 of 71 pings, rerouted via R2.

IF ASKED Q: Does RUDP run inside Packet Tracer? A: No, it cannot run C. Packet Tracer validates the network design; protocol is validated on one machine plus Wireshark.

14. Packet Tracer results

Time on this slide: 0:40

Packet Tracer results: routes, failover, ACL



OSPF routes on R1

```
Link failure: 70/71 pings,
reroute
```

ACL: student blocked, staff allowed

- Server LAN metric 2 via R3, branch LAN metric 3 via R2
- Shutting R1 Gi0/2: traffic reroutes via R2, only 1 of 71 pings lost
- Student PC gets "destination unreachable" to staff; server still reachable (TTL 126)

SAY

- Left: OSPF routes on R1 with the metrics. Middle: the link-failure test, 70 of 71 pings. Right: the ACL, where the student gets destination unreachable to staff but the server is still reachable (TTL 126).

IF ASKED If asked to show it live: have the .pkt file open and follow Demo_Steps.txt.

15. Smart-RUDP design

Time on this slide: 0:50

Smart-RUDP: a Q-learning window controller

State (80)

RTT ratio x loss fraction x window size, sampled once per RTT.

Action (5)

Multiply cwnd by 0.5, 0.85, 1.0, 1.15 or 1.5.

Reward

Throughput gained, minus delay growth, minus retransmissions.

Same Go-Back-N reliability as the baseline; only the window policy changes. Paths emulated in-program: 5G-like, lossy wireless, satellite-like.

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SAY

- Q-learning with 80 states (RTT ratio, loss fraction, window size), sampled once per RTT.
- Five actions: multiply the window by 0.5, 0.85, 1.0, 1.15 or 1.5.
- Reward: throughput gained, minus delay growth, minus retransmissions. Same Go-Back-N reliability; only the window policy changes. Paths are emulated in the program: 5G-like, lossy wireless, satellite-like.

IF ASKED Q: Is it really learning? A: Yes, 40 training episodes; the learning-curve figure shows it. It is small and tabular with limited training.

16. The AI agent in our code

Time on this slide: 1:00

The AI agent in our code

```
src/rl_cc.h (agent, simplified)
// state: 4 x 4 x 5 = 80 buckets
s = (rtt_bucket*4 + loss_bucket)*5 + cwnd_bucket;

// choose: explore with prob. eps, else best
int rl_choose(a, s) {
    if (rand01() < a->eps) return rand() % 5;
    return argmax(a->q[s]); // 5 actions
}

// learn: the Q-learning update
void rl_update(a, s, act, r, s2) {
    target = r + a->gamma * max(a->q[s2]);
    a->q[s][act] += a->alpha * (target - a->q[s][act]);
}
```

```
src/smart_client.c (decision loop, simplified)
// once per RTT, in smart_client.c
ratio = rtt / base_rtt; lossf = losses / sent;

reward = sqrt(throughput_norm)
        - 0.5 * max(0, ratio - 1) // delay growth
        - 4.0 * lossf;           // loss

rl_update(&ag, prev_s, prev_a, reward, s2);
act = rl_choose(&ag, s2);
cwnd *= {0.5, 0.85, 1.0, 1.15, 1.5}[act];
```

Tabular Q-learning; alpha 0.15, gamma 0.9. Chat version: web/cc.py, trained in web/train.py (results/q_chat.txt). Deep version: web/dqn.py, a small neural network (5-32-32-5) trained by DQN in web/train_deep.py.

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SAY

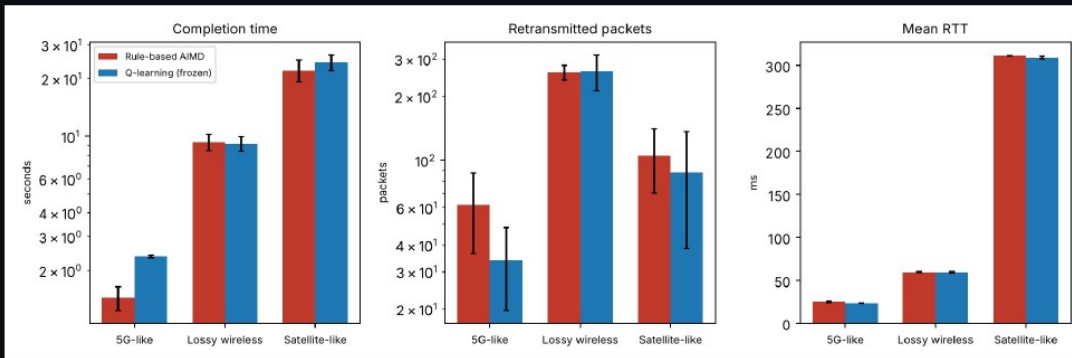
- This is the agent itself. Left: rl_cc.h, 73 lines of C. The state is 80 buckets of RTT ratio, loss and window size.
- choose picks one of five window multipliers, usually the best, sometimes random so it keeps exploring.
- update is the Q-learning rule: move the value of the action we took toward the reward plus the best value of the next state.
- Right: the loop in smart_client.c. Once per RTT it computes the reward, throughput minus delay growth minus loss, updates the table, picks the next action and scales the window.
- The code on the slide is simplified; the real files are in the repo.
- We also built a deep version: web/dqn.py is a small neural network, two hidden layers of 32, trained with DQN in the same simulator. It reads continuous inputs instead of buckets.

IF ASKED Q: Is this deep RL? A: The C agent is tabular. The chat also has a deep agent, a small DQN in numpy (5-32-32-5). Q: Does deep beat the table? A: Modestly: 3.20 s vs 3.73 s mean on 48 unseen paths, faster on 19, slower on 6. Q: Any problem? A: Training is seed-sensitive: 1 of 4 seeds failed, so we select on validation paths. It is simulated and not in the C code. Q: Where is it trained? A: web/train.py for the chat, the smart client trains online for the main experiments; table in results/q_chat.txt.

17. Smart-RUDP result

Time on this slide: 0:50

Smart-RUDP vs AIMD: honest result



- 6 evaluation runs per scenario after 40 training episodes
- Not faster: +63% time on 5G-like, +10% on satellite, tie on lossy
- Fewer retransmissions on 5G-like and satellite, but within run-to-run noise
- Earlier training rounds were much worse; reported run is the final configuration

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SAY

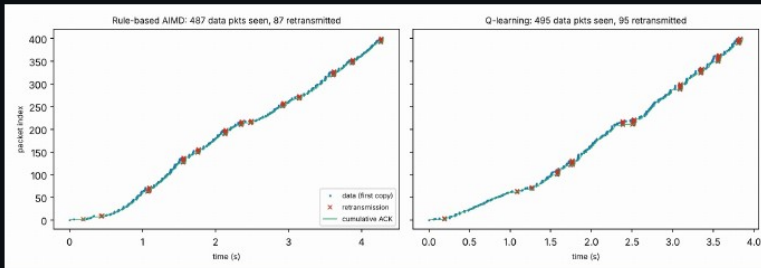
- Be honest: it did not beat AIMD. Plus 63 percent time on 5G-like, plus 10 percent on satellite, a tie on the lossy path.
- It used somewhat fewer retransmissions on two paths, within run-to-run noise.
- Earlier training rounds were much worse; this is the final configuration. The finding is that a simple learner does not beat a tuned rule.
- In the chat, retrained for bursty random-loss traffic, it does beat AIMD on most simulated paths; mention that on slide 19.

IF ASKED Q: Then what was the point? A: The experiment. We tested whether a learned window helps and report the answer, including the negative result.

18. Wireshark traces

Time on this slide: 0:40

Wireshark: RUDP dissector and traces



- rudp.lua decodes seq, ack, flags, checksum, length
- Filters: `smartrudp.flags.data == 1`, `smartrudp.seq == N`
- 400 packets, lossy path: 87 (AIMD) and 95 (Smart) repeated data segments seen at the receiver
- Trace exported by the program, topology addresses, handshake not included

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SAY

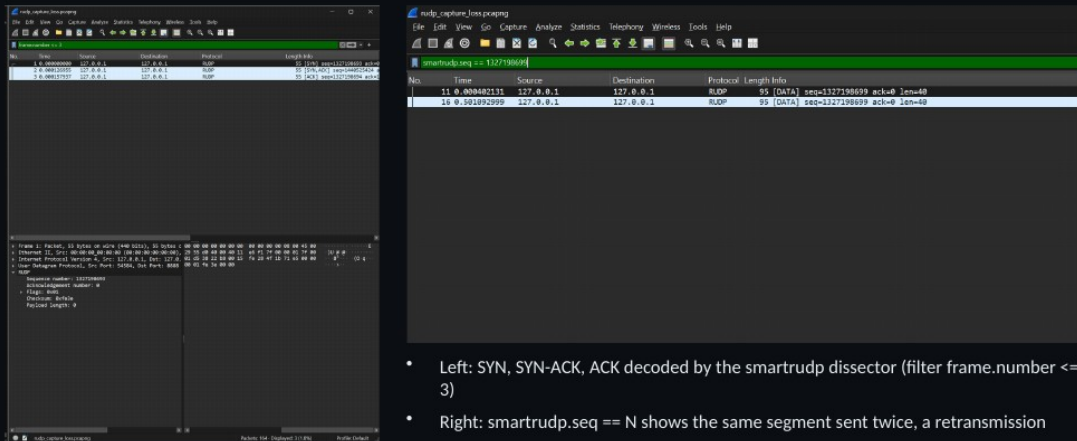
- We wrote a Lua dissector, named smartrudp, that decodes sequence, ack, flags, checksum and length.
- Useful filters: `smartrudp.flags.data == 1` and `smartrudp.seq == N`.
- In a 400-packet lossy run the receiver saw 87 (AIMD) and 95 (Smart) repeated data segments.
- The program's exported trace uses topology addresses and has no handshake.

IF ASKED Q: Why no handshake in the export? A: The export builds synthetic packets. The real tshark capture on the next slide shows the handshake.

19. Wireshark real capture

Time on this slide: 0:50

Wireshark: real capture



SAY

- Left: the first three packets, SYN, SYN-ACK and ACK, decoded by our dissector (filter frame.number <= 3).
- Right: filter smartруд.seq == N shows the same segment twice, a retransmission.

IF ASKED If Wireshark is open live: Analyze, Decode As, UDP 8888 = RUDP, then use these two filters.

20. RUDP Chat demo

Time on this slide: 1:00

Demo: RUDP Chat over the internet

RUDP Chat • connected to 127.0.0.1:9202

you are Harsh

#52 (04:10) → delivered

Harsh: test message 52/60

#53 (04:10) → delivered

Harsh: test message 53/60

#54 (04:10) → delivered

Harsh: test message 54/60

#55 (04:10) → delivered

Harsh: test message 55/60

#56 (04:10) → delivered

Harsh: test message 56/60

#57 (04:10) → delivered

Harsh: test message 57/60

#58 (04:10) → delivered

Harsh: test message 58/60

#59 (04:10) → delivered

Harsh: test message 59/60

#60 (04:10) → delivered

Harsh: test message 60/60

PACKET LOSS TO SIMULATE: 10%

450 39 388

packets sent lost resent

WINDOW CONTROLLER

Smart-RUDP (Q-learning)

In flight: 0 / window: 3 cwnd: 3.4 queued: 0 last action: x1.0

Send 60 test messages

60 msgs in 1.3 s (smart)

PROTOCOL LOG

04:18:25 3 duplicate ACKs: fast retransmit from #58

04:18:35 3 duplicate ACKs: fast retransmit from #58

04:18:35 network last message #58 (retransmission)

04:18:35 3 duplicate ACKs: fast retransmit from #58

04:18:35 network last message #58 (retransmission)

04:18:35 3 duplicate ACKs: fast retransmit from #58

04:18:35 stress test done: 60 messages in 1.3 s (smart)

04:18:35 smart: rtt x1.0, loss 0%, cwnd → 3.4 (x1.0)

Type a message Send

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SAY

- The protocol in use: each message is one numbered, checksummed RUDP segment, ACKed in order.
- There is a terminal version and a browser version; two devices were linked over Tailscale, laptop and phone.
- Drag the loss slider: lost messages are resent and still arrive in order. The right panel shows sent, lost and resent counts and the log.
- There is a controller switch: fixed window, AIMD, or Smart-RUDP. For the chat we retrained Smart-RUDP on simulated paths: it was faster than AIMD on 5 of 6 held-out paths and equal on 1, because AIMD resets to one packet on random loss. Say it is simulated, and that the agent may not shrink the window without a congestion signal. Stress test: 48 random paths the agent never trained on, 3 seeds each. Mean delivery time was 3.73 seconds for Smart-RUDP against 7.16 for AIMD. Smart-RUDP was faster on 27 paths, slower on 5, within 5 percent on 16. So on our data, in simulation, it beats AIMD in the chat setting. There is also a deep neural-network option in the controller dropdown; it gave 3.20 seconds mean on the stress paths against 3.73 for the table.

IF ASKED Q: Is that a fair win? A: Both start from the same window of 10; held-out paths and seeds; honest limits: simulated, and without the no-shrink rule the agent lost. Q: Did it win in Section 8? A: No, that result stands. Q: Where does it lose? A: Tight bottleneck queues and very high loss; the fixed window of 8 is hard to beat there. A broader-trained agent did worse, so we kept the first.

21. More evidence: baselines, SACK, reordering

Time on this slide: 1:00

More evidence: baselines, SACK, reordering

48 paths, original simulator	Mean	Median
Plain UDP	68% delivered	-
TCP model (selective ACK)	4.56 s	1.48 s
RUDP fixed window	3.88 s	2.00 s
RUDP AIMD	7.16 s	2.72 s
Smart-RUDP, table	3.73 s	1.99 s
Smart-RUDP, deep	3.20 s	1.79 s

Caveat: this simulator allowed a dup-ACK retransmission storm. With the realistic rule, stress means are AIMD 7.75 s, neural 7.27 s: about level. The TCP row is a model, not kernel TCP.

- Selective ACK (simulated): AIMD mean 7.16 s to 4.90 s, median 2.72 s to 1.44 s; faster on 34 of 48 paths. Go-Back-N is the main weakness.
- Reordering (20 ms jitter): AIMD 0.82 s to 6.79 s; deep agent 0.93 s; SACK removes most of the damage.
- 20-run intervals for the C experiments: Smart slower on 5G-like (+0.92 s) and satellite-like (+6.3 s), tie on lossy.

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SAY

- The abstract also promised TCP and plain UDP baselines, so we ran them on the same 48 unseen simulated paths.
- Plain UDP delivered only 68 percent of the messages, so reliability is needed. Our TCP model, with selective acknowledgments, has the lowest median at 1.48 seconds.
- Smart-RUDP deep is at 3.20 seconds mean and is about even with the TCP model path by path: faster on 22, slower on 23.
- The gap to TCP is mostly Go-Back-N. A selective-ack version of our transport cut AIMD from 7.16 to 4.90 seconds in simulation.
- Reordering hurts Go-Back-N badly: AIMD goes from 0.82 to 6.79 seconds at 20 milliseconds of jitter. The learned agents cope better and selective ack removes most of the damage.
- For the C experiments we repeated 20 runs per mode: Smart-RUDP is still slower on 5G and satellite and ties on lossy.
- Be upfront: our simulator originally allowed a duplicate-ACK retransmission storm that flatters aggressive controllers. With the realistic rule the neural agent is about level with AIMD in simulation (7.27 s against 7.75 s). The C results are the evidence we rely on.

IF ASKED Q: Is the TCP row real TCP? A: No, it is a model with the same window cap; real kernel TCP needs netem on Linux, and we provide the script. Q: Why did you not implement SACK in C? A: Time; the simulation shows it is the biggest win and it is the first item of future work. Q: Is 20 runs enough? A: The intervals exclude zero for 5G and satellite and include zero for lossy. Did you check it is not a simulator artefact? Yes: with the realistic rule the simulated gain mostly disappears, so we lean on the C results, which use that rule and equal window caps.

22. Neural agent in the C transport

Time on this slide: 1:00

Neural agent in the C transport

1,200 packets, C transport	AIMD	Table (r12)	Neural
5G-like	1.38 s	2.64 s	1.18 s
Lossy wireless	9.13 s	7.54 s	6.63 s
Satellite-like	17.9 s	38.1 s	13.1 s

- Neural agent beats AIMD on all three paths: 15%, 27%, 24% faster (95% intervals exclude zero)
- Table agent does not generalise: slower on 5G-like and satellite-like
- Selective ACK in C: about 10x fewer retransmissions, no significant time change here
- Caveats: trained in a simulator; emulated paths (not real networks); agent retransmits more on random loss

15 runs per cell (satellite: 8), 95% intervals. Equal-cap check (AIMD capped at 32, 10 runs): neural still wins by 19%, 29%, 28%, all significant (eval_ci4.sh).

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SAY

- We then put the neural agent into the C transport, the same code as Section 8, and tested it on the same three emulated paths with 15 runs each.
- AIMD and the agents all start from a window of 10. The neural agent was faster than AIMD on all three: 15 percent on 5G-like, 27 percent on lossy, 24 percent on satellite-like. The intervals exclude zero.
- The retrained table did not carry over: it was slower on 5G and satellite. A network generalises to unseen paths where a table does not.
- We also added selective acknowledgments to the C code: about ten times fewer retransmissions, but no significant change in completion time on these emulated paths.
- We also reran it with AIMD capped at 32 like the agent: the neural agent still wins by 19, 29 and 28 percent, all significant.

IF ASKED Q: So Smart-RUDP now beats AIMD? A: The neural version does, in the C transport on emulated paths, trained in a simulator; the original table agent in Section 8 did not. Q: Is it fair? A: Same start window; AIMD keeps a cap of 128 while the agents are capped at 32; the agent retransmits more on random loss. Q: Why no gain from SACK? A: Those paths are limited by the bottleneck rate, so extra retransmissions cost little; the simulator, with different paths, showed a large gain.

23. Fairness, ablations, chat fixes

Time on this slide: 0:50

Fairness, ablations, chat fixes

Two flows, one bottleneck	Share of A	Jain
AIMD vs AIMD (long path)	51%	0.999
Deep vs deep (long path)	59%	0.969
Deep vs AIMD (no-loss path)	44%	0.987
Deep vs AIMD (long path)	74%	0.819
Fixed 8 vs AIMD (long path)	72%	0.843

Simulator, 20 seeds; ablation on 48 stress paths. Code: [web/fairness_test.py](#), [web/ablate.py](#).

- Fair on short paths; takes 74% against AIMD on the 120 ms path, like a fixed window
- Ablation: the no-shrink mask is what matters (6.54 s without it, about 3.3 s with it; AIMD 7.16 s)
- Other parts (delay penalty, last action, throughput input) showed no measurable value
- Same recipe gave 3.2 to 7.4 s across seeds: seed sensitivity is a real weakness
- Found and fixed a dup-ACK retransmission storm in the chat gateway; added a live dashboard and an adaptive RTO

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SAY

- We also tested two things people ask about. Fairness: two flows through one bottleneck. The neural agent is fair against itself and on short paths, but takes 74 percent against AIMD on the long 120 ms path, about like a fixed window.
- Ablation: we retrained the agent with one piece removed. Only the no-shrink mask matters; without it the agent is no better than AIMD. The other pieces showed no measurable value.
- The same recipe also gave very different results by seed, between 3.2 and 7.4 seconds, so we pick the seed on validation paths.
- While building the dashboard we found a retransmission storm in the web gateway and fixed it; there is also an adaptive timeout option now.

IF ASKED Is the agent fair? Mostly, but not on the long path against AIMD (74% share, Jain 0.82). Fairness was only tested in the simulator, not in C.

24. Safety net: hand over to AIMD

Time on this slide: 0:50

Safety net: hand over to AIMD

C transport, seconds	AIMD 32	Neural	Hybrid
5G-like	1.48	1.17	1.23
Lossy 2%	9.04	7.53	7.57
Satellite-like	19.7	13.4	13.7
Hostile, 10% loss	11.1	84.9	13.9

10 runs per cell (hostile: 5 runs, 300 packets), 95% intervals. Flag: --hybrid 0.10. Code: eval_ci6.sh.

- Problem found: at 10% loss the neural agent is 7x slower than AIMD
- Fix: while a decision interval loses 10% or more, AIMD drives the window; after 3 clean intervals the agent resumes
- Keeps the wins on the three benign paths, and the hostile path is no longer significantly different from AIMD
- A timeout cut (x0.25) was tried and dropped: it helped the hostile path but hurt the 2% path (24 s vs 9 s)
- Honest summary: the agent helps on clean-to-moderate paths; it needs a classical safety net elsewhere

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SAY

- While checking our own result we found where the neural agent fails: on a hostile path with 10 percent loss it needed about 85 seconds where AIMD needed 11.
- The fix is a hand-over: when a decision interval loses 10 percent or more of its packets, plain AIMD controls the window; after three clean intervals the agent takes over again.
- With that, the three benign paths keep the agent's advantage and the hostile path is back near AIMD at about 14 seconds.
- We also tried cutting the window on every timeout; it fixed the hostile case but made the 2 percent path much worse, so we dropped it. So the learned agent is useful on clean to moderate paths and needs a classical safety net elsewhere.

IF ASKED Is the hybrid still 'AI'? Yes on clean and moderate paths, where the neural network decides; on heavy loss it deliberately defers to AIMD. It is a rule we added, not something the network learned.

25. Selective Repeat is now the default

Time on this slide: 0:50

Selective Repeat is now the default

C transport, seconds	AIMD 32	Neural	Gain
5G-like	1.27	1.16	9%
Lossy 2%	7.99	4.29	46%
Satellite-like	25.7	12.9	50%
Hostile, 10% loss	7.3	8.1	tie

10 runs per cell (hostile: 5), 95% intervals, all gains significant except the hostile row.
Code: eval_ci7.sh.

- Receiver buffers out-of-order packets (64 window) and reports them in a bitmap in every ACK
- Sender: one timer per packet; resends only the missing ones
- Retransmissions drop about 10x; web chat 200 messages: 6.4 s vs 11.9 s (AIMD)
- The neural agent now wins by more, and the 10% loss failure was mostly a Go-Back-N effect
- Go-Back-N kept for comparison: --gbn / RUDP_GBN

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SAY

- We swapped Go-Back-N for Selective Repeat as the default. The receiver now keeps out-of-order packets and tells the sender exactly which ones it holds in a bitmap inside every ACK; the sender has one timer per packet and resends only what is missing.
- The result: about ten times fewer retransmissions, and the neural agent's lead over AIMD grows to 46 percent on the lossy path and 50 percent on the satellite path.
- The 10 percent loss failure we found earlier was mostly a Go-Back-N effect: with Selective Repeat the agent ties AIMD there.
- Go-Back-N is still in the code behind a flag, and the earlier tables in the report were measured with it and are labelled that way.

IF ASKED Is it textbook Selective Repeat? Nearly: per-packet timers and a buffering receiver, but we use a cumulative ACK plus a 64-bit bitmap instead of one ACK per packet, and the window is capped at 60.

26. Learnings and future work

Time on this slide: 0:50

Learnings and future work

Learned

Reliability is a set of simple rules working together.
Congestion control matters as much as retransmission.
Testing tools fail in surprising ways.

Next

Train the neural agent in C / on real netem
Selective ACK where throughput limits the path
Kernel netem and real multi-machine tests

Limits: Go-Back-N wastes bandwidth and suffers under reordering; C tests use an emulated path; the tabular agent did not beat AIMD in C.

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SAY

- Learned: reliability is a set of simple rules working together; congestion control matters as much as retransmission; testing tools fail in surprising ways.
- Next: Selective Repeat with SACK, a stronger learning agent, adaptive timeout, and tests on a real multi-machine network.
- Name the limits before she does: Go-Back-N, loopback-only protocol tests, Smart-RUDP not beating AIMD.

IF ASKED End on the limits and the fix order: SACK, adaptive timeout, real-network tests.

27. Thank you

Time on this slide: 0:15

Thank you

`github.com/Capsturrrr/rudp`

Harshavardhan Mallela Rohit Arya Shloak Sinha

SAY

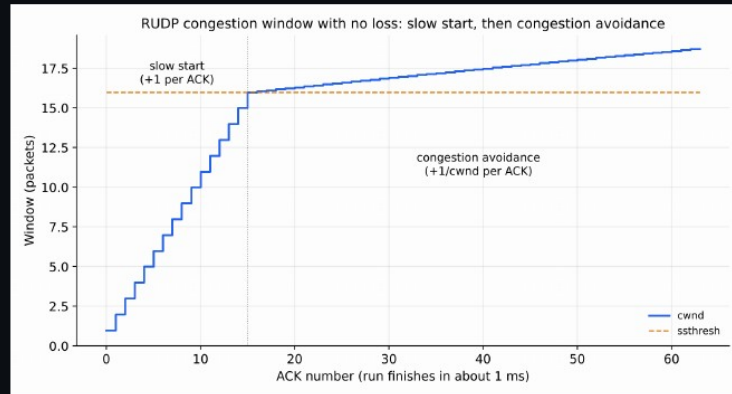
- Thank you. The code, report and captures are at github.com/Capsturrrr/rudp. Happy to take questions.

IF ASKED Pause and smile; hand over to questions.

28. Appendix: slow start

Time on this slide: -

Appendix: slow start and congestion avoidance



- cwnd starts at 1
- +1 per ACK until ssthresh = 16
- Then +1/cwnd per ACK
- 62 packets, final cwnd 18.71

Appendix Smart-RUDP

SAY

- Only if asked: the window starts at 1, adds 1 per ACK until the threshold 16, then 1/window per ACK. In the 62-packet run the final window was 18.71.

IF ASKED Use this slide to answer 'show me congestion avoidance'.

Hard questions and short answers (1 of 2)

Is this a lab exercise?

A lab sends datagrams. We built reliability and congestion control on top, measured them, and used them in a real chat and a routed network.

Why not TCP?

TCP is in the kernel and cannot be changed. Building our own makes every mechanism visible. QUIC and HTTP/3 do the same.

Did Smart-RUDP win?

Experiments: tied on one path, slower on two, and we say so. Chat, retrained on simulated paths: faster than AIMD on 5 of 6 held-out paths, equal on 1, because AIMD resets to 1 on every timeout. Deep RL: a small DQN in the chat gateway gave 3.20 s mean on the same stress paths (training is seed-sensitive; chosen on validation paths). Stress test on 48 unseen random paths: mean 3.73 s Smart vs 7.16 s AIMD; faster on 27, slower on 5, within 5% on 16. Simulated;

Why Go-Back-N?

We started with it because it is simple to build and explain, then replaced it as the default with Selective Repeat: the receiver buffers out-of-order packets and reports them in an ACK bitmap, and the sender resends only what is missing. Go-Back-N stays behind a flag for comparison.

Tests ran on one machine?

Yes, with emulated delay, loss and jitter. They do not prove internet performance; the report says so. Chat between laptop and phone is a small real-network check.

Hard questions and short answers (2 of 2)

What does Packet Tracer prove?

The network design: OSPF, VLANs, ACL, 70/71 failover. The protocol is validated separately with emulation and Wireshark.

Peer disappears?

Sender keeps retransmitting on timeout. A connection abort after repeated timeouts is not built; it is the fix we would add.

Does the checksum catch errors?

16-bit one's-complement sum, like UDP/TCP. Corrupted packets fail and are retransmitted. It cannot catch every error pattern.

Why no handshake in the exported trace?

The export builds synthetic packets. The real tshark capture shows SYN, SYN-ACK, ACK.

Is Q-learning really learning?

80 states, 5 actions, 40 episodes; the learning-curve figure shows it. Small and tabular, with limited training.